

# LCDSirPlus 0.3.0 Instruction Manual

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**Draft / Unpublished:** LCDSirPlus 0.3.0 is not published. The package described here is planned, and final package testing remains pending.

LCDSirPlus runs on Windows 11 x64. A Logitech G13 is needed only for the physical LCD and buttons. The virtual preview works without one.

Start here: [verify a download](#), [install or extract it](#), then follow [first start](#).

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## Read display messages

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- **N/A** : the required device or software is not configured, connected, or returning a usable reading. Check the requirement in the button option guide.
- **STALE** : the displayed value is old because updates stopped. Do not trust it; check the required device or software.
- **OFFLINE** : a supported device is disconnected or turned off.
- **DISABLED** : the feature or safe mode turned the data source off.
- **IDLE** : no active PresentMon game session.
- **CLEAR** : no alert or no available temporary button option.
- **DOWN** : one or more tracked sources are disabled, unconfigured, disconnected, or unavailable. Optional sources you do not use can count as down, so this is not necessarily a failure.
- **NO\_DONGLE** : no supported SteelSeries wireless receiver was found.
- **UNKNOWN** : Windows did not provide a usable system battery state.
- **AC\_ONLY** : AC power is connected and Windows reports no battery.
- **NO\_BAT** : AC power is disconnected and Windows reports no battery.
- **CHG** : the system battery is charging; a percentage follows.
- **AC** : the system is on AC power; a battery percentage follows.
- **BAT** : the system is on battery power; a percentage follows.

- **OK** : all sources included in the current status summary are ready. In safe mode, this means the restricted safe-mode dashboard is healthy.
- **WIRED** : an XInput controller is connected by cable, so no battery percentage is available.
- **MUTED** : the default Windows microphone is muted.
- **LIVE** : the default Windows microphone is not muted.

In safe mode, optional panels can show inactive states such as **DISABLED**, **OFFLINE**, **N/A**, or **CLEAR**. These states describe the restricted dashboard; they do not mean safe mode itself failed.

## Verify a download

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The planned LCDSirPlus packages are unsigned. When they are published, open Terminal in the folder containing the package and `LCDSirPlus-0.3.0-SHA256SUMS.txt`, then run the matching command:

```
Get-FileHash .\LCDSirPlus-0.3.0-win-x64-setup.exe -Algorithm SHA256
Get-FileHash .\LCDSirPlus-0.3.0-win-x64-portable.zip -Algorithm SHA256
```

The displayed hash must exactly match the line for the same filename in the checksum file. Do not run or extract the package if the filename is missing or the hash differs. The planned ZIP also has `PACKAGE-MANIFEST.txt`, which lists every file's checksum and size. PresentMon has its own Intel signature.

## Install or run portable

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### Install for the current user or all users

Exit LCDSirPlus, then run `LCDSirPlus-0.3.0-win-x64-setup.exe`.

- Current-user program files go to `%LOCALAPPDATA%\Programs\LCDSirPlus`.
- All-users program files go to `%ProgramFiles%\LCDSirPlus` after approval.
- Start Menu shortcuts for LCDSirPlus, this manual, and the security guide are selected by default, as is **Start LCDSirPlus when I sign in**.
- The desktop shortcut is optional.

The planned install also includes `SECURITY.md`, `RELEASE-NOTES.md`, and `modules.md` in the program folder.

An all-users install makes program files available to every user. The selected sign-in startup task belongs only to the Windows account that ran setup. Setup does not automatically add a startup task for other users.

The first installed start creates this user-owned file only if it is missing:

```
%LOCALAPPDATA%\LCDSirPlus\Config\lcdsirplus.txt
```

Updates and repairs do not replace it.

### Run the portable package

1. Extract the ZIP to a folder on a local drive.
2. Open the folder in File Explorer.
3. Right-click empty space and select **Open in Terminal**.
4. Run `.\LCDSirPlus.exe --preview`.

Keep `LCDSirPlus.exe`, `PresentMon.exe`, `lcdsirplus.txt`, `SECURITY.md`, `RELEASE-NOTES.md`, `modules.md`, and the `docs` folder together. Portable use creates no installer-owned shortcuts.

## Run commands from the correct folder

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Commands beginning with `.\LCDSirPlus.exe` must run in the folder containing that executable. Use **Open in Terminal** there first.

For either an installed or portable copy, open Terminal in the folder that contains `LCDSirPlus.exe`. All command examples below are package-relative and use `.\LCDSirPlus.exe`.

## Start LCDSirPlus the first time

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### Installed copy

Open **Start**, find **LCDSirPlus**, and select it.

### Portable copy

Double-click `LCDSirPlus.exe`, or run it from the open Terminal.

Expected result:

- The LCDSirPlus tray icon appears.
- Without a G13, the preview appears automatically.
- With a working physical G13 connection, the preview normally starts hidden.
- Left-click the tray icon to show or hide the preview.
- Right-click it for **Toggle Preview** and **Exit**.
- Closing the preview window exits LCDSirPlus.
- Showing or clicking the preview does not take keyboard focus from the active game or app.

Only one normal LCDSirPlus process can run in one Windows session.

## Open and edit settings

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### Installed copy

1. Press **Win+R**.
2. Enter `%LOCALAPPDATA%\LCDSirPlus\Config` and press Enter.
3. Right-click `lcdsirplus.txt`.
4. Select **Open with**, then **Notepad**.

### Portable copy

Open `lcdsirplus.txt` beside `LCDSirPlus.exe` in Notepad.

Save after editing. Valid changes apply automatically. An invalid change is rejected, and the last valid settings remain active. Validate with:

```
.\LCDSirPlus.exe --validate-config
```

An explicit file can be checked with:

```
.\LCDSirPlus.exe --validate-config --config .\lcdsirplus.txt
```

The format uses one setting followed by values. # starts a comment. Quote a value that contains spaces. See CONFIGURATION.md for every key, default, range, and include-file rule.

## Choose layouts and button options

Set one main layout:

```
main_display 1
```

- 1 : CPU/RAM and GPU/VRAM halves.
- 2 : CPU/RAM, GPU/VRAM, and outgoing/incoming thirds.
- 3 : CPU/RAM and incoming/outgoing halves.

All three keep the date/time header and four lower button slots. Set the network ceilings to your internet plan's downstream and upstream Mbps; LCDSirPlus does not auto-detect them. For a 1000-down/40-up plan:

```
network_graph_ceiling_download_mbps 1000
network_graph_ceiling_upload_mbps 40
```

Download is inbound/IN: NET\_IN\_GRAPH, the top half of NET\_GRAPH, layout 2 IN, and layout 3 NET IN. Upload is outbound/OUT: NET\_OUT\_GRAPH, the bottom half of NET\_GRAPH, layout 2 OUT, and layout 3 NET OUT. Each direction reaches full utilization independently at its own ceiling. Numeric NET\_IN, NET\_OUT, and NET\_BOTH readings do not change.

Old network\_graph\_ceiling\_mbps lines remain active and set both directions at that point in file/include order. A later directional line replaces only its direction; a later legacy line replaces both. Use the two directional lines for new configurations. Both directional defaults are 1000, and all three spellings accept finite values from 1 through 100000.

The header and CLOCK use your current Windows Short time preference, including the Windows 12/24-hour selection (for example, 2:25 PM or 14:25). Changes to that Windows preference are picked up while LCDSirPlus runs.

Each slot is an ordered list for one physical button and preview area:

```
slot_0 HEADSET_BATTERY CPU_TEMP CONTROLLER_BATTERY
slot_1 FPS_CURRENT NET_BOTH NET_GRAPH FRAME_TIME SESSION_TIME
slot_2 PROC_HANG FPS_1LOW DISK_IO DISK_IO_GRAPH GPU_TEMP PING JITTER AUDIO
slot_3 BOTTLENECK THERMALS POWER_LIMIT
```

A short physical press moves forward. Preview left-click moves forward and right-click moves backward. Button 4 first acknowledges the highest active alert. PROC\_HANG can appear only once across all slots.

### Button option guide

The quick reference in modules.md mirrors this table exactly. In an installed or portable package, open modules.md beside LCDSirPlus.exe.

| Option             | Description                                                                                                            |
|--------------------|------------------------------------------------------------------------------------------------------------------------|
| HEADSET_BATTERY    | Shows battery and connection. What you need: A supported SteelSeries Arctis/GameBuds wireless USB receiver.            |
| CONTROLLER_BATTERY | Shows controller battery or WIRED. What you need: A controller that Windows recognizes as an Xbox (XInput) controller. |

| Option          | Description                                                                                                                                                                                                            |
|-----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| FPS_CURRENT     | Shows current FPS. What you need: PresentMon, planned for the 0.3.0 package.                                                                                                                                           |
| FPS_1LOW        | Shows common FPS slowdowns. What you need: PresentMon, planned for the 0.3.0 package.                                                                                                                                  |
| FPS_01LOW       | Shows rarer severe FPS slowdowns. What you need: PresentMon, planned for the 0.3.0 package.                                                                                                                            |
| FRAME_TIME      | Shows frame time and a 30-second graph. What you need: PresentMon, planned for the 0.3.0 package.                                                                                                                      |
| CPU_TEMP        | Shows CPU temperature. What you need: HWiNFO or LibreHardwareMonitor, installed and set up separately.                                                                                                                 |
| GPU_TEMP        | Shows GPU temperature. What you need: An NVIDIA/AMD graphics driver, or LibreHardwareMonitor set up separately.                                                                                                        |
| NET_IN          | Shows the current incoming rate. What you need: Built into Windows and LCDSirPlus; no extra software.                                                                                                                  |
| NET_OUT         | Shows the current outgoing rate. What you need: Built into Windows and LCDSirPlus; no extra software.                                                                                                                  |
| NET_BOTH        | Shows incoming plus outgoing. What you need: Built into Windows and LCDSirPlus; no extra software.                                                                                                                     |
| NET_IN_GRAPH    | Shows incoming traffic for 30 seconds. What you need: Built into Windows and LCDSirPlus; no extra software.                                                                                                            |
| NET_OUT_GRAPH   | Shows outgoing traffic for 30 seconds. What you need: Built into Windows and LCDSirPlus; no extra software.                                                                                                            |
| NET_GRAPH       | Shows both network directions for 30 seconds. What you need: Built into Windows and LCDSirPlus; no extra software.                                                                                                     |
| PING            | Shows the latest network delay. What you need: The optional network probe, enabled and configured separately.                                                                                                          |
| JITTER          | Shows network delay variation. What you need: The optional network probe, enabled and configured separately.                                                                                                           |
| PACKET_LOSS     | Shows the percentage of failed checks. What you need: The optional network probe, enabled and configured separately.                                                                                                   |
| MIC_STATUS      | Shows LIVE or MUTED. What you need: Built into Windows and LCDSirPlus; no extra software.                                                                                                                              |
| AUDIO           | Shows default output volume. What you need: Built into Windows and LCDSirPlus; no extra software.                                                                                                                      |
| SESSION_TIME    | Shows active game-session time. What you need: PresentMon, planned for the 0.3.0 package.                                                                                                                              |
| SESSION_SUMMARY | Shows session time and stutters. What you need: PresentMon, planned for the 0.3.0 package.                                                                                                                             |
| CLOCK           | Shows time using the Windows Short time format. What you need: Built into Windows and LCDSirPlus; no extra software.                                                                                                   |
| GAME_NAME       | Shows the selected game or app filename. What you need: PresentMon, planned for the 0.3.0 package. A filename may appear under either persist setting; set <code>presentmon_enabled 0</code> to prevent this exposure. |
| ALERTS          | Shows unacknowledged active alerts or CLEAR. What you need: Built into Windows and LCDSirPlus; no extra software.                                                                                                      |
| PROVIDER_STATUS | Shows unavailable tracked data sources or OK. What you need: Built into Windows and LCDSirPlus; no extra software.                                                                                                     |
| CPU_LOAD        | Shows total CPU use. What you need: Built into Windows and LCDSirPlus; no extra software.                                                                                                                              |
| RAM_USAGE       | Shows memory use and capacity. What you need: Built into Windows and LCDSirPlus; no extra software.                                                                                                                    |
| GPU_LOAD        | Shows GPU use. What you need: An NVIDIA or AMD graphics driver.                                                                                                                                                        |
| VRAM_USAGE      | Shows video memory use and capacity. What you need: An NVIDIA or AMD graphics driver.                                                                                                                                  |
| CPU_CACHE_TEMP  | Shows N/A. What you need: Not available in the draft 0.3.0 build.                                                                                                                                                      |
| CPU_FREQ_TEMP   | Shows N/A. What you need: Not available in the draft 0.3.0 build.                                                                                                                                                      |
| CPU_LOAD_GRAPH  | Shows CPU use for 30 seconds. What you need: Built into Windows and LCDSirPlus; no extra software.                                                                                                                     |
| GPU_LOAD_GRAPH  | Shows GPU use for 30 seconds. What you need: An NVIDIA or AMD graphics driver.                                                                                                                                         |
| CPU_TEMP_GRAPH  | Shows CPU temperature for 30 seconds. What you need: HWiNFO or LibreHardwareMonitor, installed and set up separately.                                                                                                  |
| GPU_TEMP_GRAPH  | Shows GPU temperature for 30 seconds. What you need: An NVIDIA/AMD graphics driver, or LibreHardwareMonitor set up separately.                                                                                         |

| Option           | Description                                                                                                                                                                                                                                       |
|------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| VRM_TEMP         | Shows motherboard power-circuit temperature. What you need: LibreHardwareMonitor with the correct VRM temperature selected.                                                                                                                       |
| CPU_FAN          | Shows CPU fan percentage. What you need: LibreHardwareMonitor with the correct CPU fan selected.                                                                                                                                                  |
| PUMP_RPM         | Shows pump percentage. What you need: LibreHardwareMonitor, set up with the matching pump reading selected.                                                                                                                                       |
| POWER_LIMIT      | Shows total system power. What you need: One total-power reading selected in HWiNFO or LibreHardwareMonitor.                                                                                                                                      |
| CPU_GPU_POWER    | Shows CPU plus GPU power. What you need: CPU power from HWiNFO or LibreHardwareMonitor, plus GPU power from the graphics driver, HWiNFO, or LibreHardwareMonitor.                                                                                 |
| CHIPSET_TEMP     | Shows chipset temperature. What you need: LibreHardwareMonitor, set up with the matching sensor selected.                                                                                                                                         |
| MOTHERBOARD_TEMP | Shows motherboard temperature. What you need: LibreHardwareMonitor, set up with the matching sensor selected.                                                                                                                                     |
| DISK_IO          | Shows total disk read and write rates. What you need: Built into Windows and LCDSirPlus; no extra software.                                                                                                                                       |
| DISK_IO_GRAPH    | Shows disk activity for 30 seconds. What you need: Built into Windows and LCDSirPlus; no extra software.                                                                                                                                          |
| RAM_DETAIL       | Shows used and total memory. What you need: Built into Windows and LCDSirPlus; no extra software.                                                                                                                                                 |
| FPS_GRAPH        | Shows FPS for 30 seconds. What you need: PresentMon, planned for the 0.3.0 package.                                                                                                                                                               |
| THERMALS         | Shows CPU and GPU temperatures. What you need: Both CPU and GPU temperature sources set up and current.                                                                                                                                           |
| CONNECTIONS      | Shows established IPv4 and IPv6 TCP connections only. What you need: Built into Windows and LCDSirPlus; no extra software.                                                                                                                        |
| NET_HEALTH       | Shows ping, jitter, and packet loss. What you need: The optional network probe, enabled and configured separately.                                                                                                                                |
| SYSTEM_BATTERY   | Shows AC, battery, and charging. What you need: Built into Windows and LCDSirPlus; no extra software.                                                                                                                                             |
| HARD_FAULTS      | Shows memory-to-disk pressure. What you need: Built into Windows and LCDSirPlus; no extra software.                                                                                                                                               |
| BOTTLENECK       | Shows CPU, GPU, RAM, or disk. With current NONE, the next eligible option appears temporarily or the slot shows CLEAR. What you need: Built-in system readings; GPU results also need a supported graphics driver.                                |
| PROC_HANG        | Shows a hung-window target. With no target, including while disabled, the next eligible option appears temporarily or the slot shows CLEAR. What you need: Not supported for end users in this draft release; leave <code>hang_enabled 0</code> . |

## Show game FPS

The planned packages include PresentMon, and FPS is enabled by default.

1. Keep `FPS_CURRENT` in a slot. The default `slot_1` already has it.
2. Start LCDSirPlus.
3. Start the game.
4. Press the second LCD button, or click the second preview slot, until current FPS appears.

Game selection is automatic. You do not maintain a game list.

The useful defaults are:

```
presentmon_enabled      1
presentmon_target_mode  presenting
presentmon_deferred     1
presentmon_persist      0
```

`presentmon_deferred 1` temporarily skips any unavailable PresentMon option for display without changing the stored choice. For example, with `slot_2 FPS_CURRENT GPU_TEMP`, a closed game can make the slot show GPU temperature; FPS returns when game data becomes available. A temporary result is never `PROC_HANG` or `BOTTLENECK`, and an all-unavailable list shows `CLEAR`. Set `presentmon_deferred 0` to keep the selected inactive panel visible instead.

`presentmon_persist 0` is the normal game preference. When a valid NVIDIA App game list is available, ordinary desktop apps are rejected. If that list is unavailable, automatic fallback may still select another app. Set it to `1` to allow desktop apps even when the game list is available. Under either persist value, the selected executable filename and FPS may appear; old readings are not preserved. If that exposure is not acceptable, set `presentmon_enabled 0`. Advanced details are in `CONFIGURATION.md`.

If Windows denies capture, add your Windows user to **Performance Log Users**, then sign out and sign in. Run elevated only to diagnose access; elevation is not the normal operating mode.

## Set up CPU temperature

CPU temperature needs either [HWiNFO](#) or [LibreHardwareMonitor](#), installed and set up separately. LCDSirPlus has no fixed minimum supported version for either tool; use a current official release and validate your setup.

### Use HWiNFO

1. Download and install HWiNFO from its official site.
2. Start HWiNFO, select **Sensors-only**, and click **Start**.
3. In the Sensors window, click the settings gear.
4. On the **General** tab, enable **Shared Memory Support**, then click **OK**.
5. In Terminal opened in the LCDSirPlus executable folder, run:

```
.\LCDSirPlus.exe --list-hwinfo-sensors
```

6. Find the CPU package/control temperature you want. Output resembles:

```
sensor="CPU [#0]" reading="CPU (Tctl/Tdie)" type=Temperature value=60 unit="C"
```

7. Copy your exact, case-sensitive labels into the installed or portable settings file:

```
hwinfo_cpu_temp_sensor "CPU [#0]"  
hwinfo_cpu_temp_reading "CPU (Tctl/Tdie)"
```

8. Save and validate.

Both settings must be present together. HWiNFO64 Free shared-memory monitoring has a 12-hour limit. For unattended use, use HWiNFO32 or an appropriately licensed HWiNFO Pro. LCDSirPlus does not start or configure HWiNFO.

### Use LibreHardwareMonitor

1. Download a current official LibreHardwareMonitor release.
2. Extract it and start `LibreHardwareMonitor.exe`.

3. In its menu, select **Options > Remote Web Server > Run**.
4. Open `http://127.0.0.1:8085/data.json` in a browser on the same computer.
5. Keep these LCDSirPlus defaults:

```
lhm_mode auto
lhm_url auto
```

6. Save and validate. Automatic CPU temperature selection may be enough.
7. If it selects the wrong temperature, copy the exact sensor path shown by `data.json`, for example:

```
lhm_cpu_temp_sensor /amdcpu/0/temperature/2
```

LCDSirPlus accepts only a local LibreHardwareMonitor address. It does not enable the web server or request a firewall exception. Do not expose this server to other computers.

`CPU_CACHE_TEMP` and `CPU_FREQ_TEMP` are not available in this release.

## Connect devices and network data

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`HEADSET_BATTERY` supports only documented SteelSeries Arctis/GameBuds wireless USB receivers. Bluetooth-only and wired models are unsupported. SteelSeries GG is not required. GameBuds shows the lower active-earbud battery, not the case. See the exact receiver list in `CONFIGURATION.md`.

`CONTROLLER_BATTERY` uses an XInput-compatible controller. `controller_index -1` chooses the first connected controller. A wired controller shows its wired state because XInput supplies no percentage.

`AUDIO` and `MIC_STATUS` use the current default Windows sound devices.

Incoming and outgoing rates are built into Windows and send no probe traffic. Ping, jitter, and packet loss use a separate optional probe. It is off by default because it sends traffic. To enable it:

```
network_probe_enabled      1
network_probe_method       auto
network_probe_target       1.1.1.1
network_probe_interval_ms  1000
network_probe_timeout_ms   1500
network_probe_window       30
```

The target must be an IP address, not a hostname. TCP can include a port. Explicit ICMP accepts IPv4 without a port. Safe mode sends no probe traffic.

## Understand PROVIDER\_STATUS

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`PROVIDER_STATUS` summarizes tracked data sources. A new setup can show `DOWN` because Discord has no application ID and the hung-window action is disabled. This is expected and is not necessarily a failure.

Setting `discord_enabled 0` or keeping `hang_enabled 0` does not remove those two entries from the summary. Other intentionally unused sources can be turned off, for example:

```
headset_enabled      0
controller_enabled   0
audio_enabled        0
gpu_provider         off
lhm_mode             off
presentmon_enabled   0
network_probe_enabled 0
```

Safe mode reports only its healthy safe-mode state, so the button option shows `OK`. Add `PROVIDER_STATUS` to any slot if you want this summary; during normal use, `DOWN` can reflect intentionally unused sources as described above.

An inactive `PROC_HANG` or `BOTTLENECK` selection temporarily shows the next eligible option without changing the stored selection. Neither option is used as the other's temporary replacement. If no eligible option is available, the slot shows `CLEAR`.

## Connect Discord

**Never put the Discord client secret in settings, command arguments, or a text file. Copy it only when needed and paste it only into the masked prompt below. Never include it in logs, screenshots, issues, or chat.**

Discord display needs Discord Desktop and your own application:

1. Open the official [Discord Developer Portal](#).
2. Select **New Application**, enter a name, and create it.
3. Open **General Information** and copy the numeric **Application ID**.
4. Put that public number in your settings:

```
discord_enabled      1
discord_client_id     123456789012345678
```

5. Add the Discord account as an application tester if Discord requires tester access for the application.
6. In the Developer Portal, open **OAuth2 > General**. Under **Client Secret**, select **Reset Secret** if Discord requires it, then copy the displayed secret only for the masked prompt below. Do not save it in a text file.
7. Start Discord Desktop under the same Windows account and select the account you want to authorize.
8. Open Terminal in the folder containing `LCDSirPlus.exe`. The following masked workflow works in

Windows PowerShell 5 and PowerShell 7:

```
$exe = ".\LCDSirPlus.exe"
try {
    $secret = Read-Host 'Discord client secret' -AsSecureString
    $env:LCDSIRPLUS_DISCORD_CLIENT_SECRET = [System.Net.NetworkCredential]::new('',
$secret).Password
    & $exe --discord-authorize
} finally {
    Remove-Item Env:\LCDSIRPLUS_DISCORD_CLIENT_SECRET -ErrorAction SilentlyContinue
    $secret = $null
}
```

9. Approve the prompt in Discord and join a voice channel. When someone speaks, their name should appear automatically on the LCD or preview.

Discord's **Public Client** switch does not remove the secret requirement for this workflow. Windows protects the secret and access records for the current Windows account. Authorize each Discord account separately.

## Disconnect Discord

1. Open Terminal in the folder containing `LCDSirPlus.exe`.
2. Set `$exe = ".\LCDSirPlus.exe"` as in step 8.
3. Remove the local credentials:

```
& $exe --discord-clear-token
```

4. Revoke the application under Discord **User Settings > Authorized Apps**.

Local deletion and remote revocation are separate; do both after suspected exposure.

## Use alerts and safe mode

Temperature and memory alerts are independent. Only current readings create an alert. `memory_warning` and `vmem_warning` default to `100` and trigger at or above that value. Button 4 acknowledges the highest active alert.

`warning 1` flashes the selected temperature button area and suppresses its full-screen temperature alert.

`warning 0` keeps the full-screen alert.

Start safe mode with:

```
.\LCDSirPlus.exe --safe-mode
```

Safe mode limits the dashboard to the clock, built-in CPU load, memory, the preview, slot cycling, and alert acknowledgement. It disables every optional dashboard data source, Discord dashboard access, network probes, startup changes, and the destructive hung-window action. The explicit `--discord-authorize` and `--discord-clear-token` commands remain available; safe mode does not silently authorize or remove credentials.

Do not enable the hung-window action in this release. Its physical behavior has not been release-tested, and there is no supported end-user test. It can terminate a program and lose unsaved work. Keep `hang_enabled 0`.

## Create diagnostics safely

---

Run from the executable folder:

```
.\LCDSirPlus.exe --diagnostics
.\LCDSirPlus.exe --diagnostics --diagnostic-dir .\diagnostics
```

Diagnostics is a bounded offline summary, not a report of active provider state. It ignores `--config` and does not start HID, optional providers, Discord, network activity, or actions. The ZIP contains only `privacy.txt`, `report.txt`, and `manifest.txt` and is capped at 1 MiB total and 128 KiB per entry. It excludes raw logs, settings, credentials, Discord content and IDs, process/window names, paths, network targets and addresses, environment values, command lines, registry values, serial numbers, arbitrary directory listings, and device paths. It records only the sizes and counts of known rotated logs, not their contents.

Review the ZIP before sharing. Use the task-specific checks in this manual, such as `--validate-config`, `--list-hwinfo-sensors`, or `--hardware-discover`, to investigate a particular failure. Review the normal log privately and separately; do not attach it without checking every line.

The normal log is `%LOCALAPPDATA%\LCDSirPlus\lcdsirplus.log`.

## Run command-line tools

---

```
LCDSirPlus.exe [--config PATH] [COMMAND]
(none)           Run the dashboard application
--validate-config Validate the configuration and exit
--list-hwinfo-sensors List usable HWiNFO sensor/reading labels and exit
--preview        Run with the virtual preview forced on
--hardware-test   Run the deterministic 10-step G13 test sequence
--hardware-discover Inspect up to 256 HID interfaces for the first exact G13 match
--diagnostics     Write a bounded offline diagnostics ZIP and exit
--discord-authorize Authorize the active Discord Desktop account
--discord-clear-token Remove all LCDSirPlus Discord credentials
--backend auto|sdk|hid|virtual Hardware test: sdk=Logitech software, hid=direct USB
--duration-secs N Duration for timed test commands (1..3600 seconds)
--safe-mode       Disable optional dashboard data sources and destructive actions
--diagnostic-dir PATH Log/diagnostic output directory
--help, -h        Print command help
--version, -v     Print version
```

Exit codes are `0` success, `1` operating failure, `2` invalid arguments or settings, `3` hardware discovery, selection, or open refusal including process-gate failures, `4` hardware-test step or shutdown failure, and `5` another LCDSirPlus instance or single-instance check failure.

## Check a Logitech G13

---

The preview verifies rendering without physical hardware. For a G13, first check HID compatibility without writing:

```
.\LCDSirPlus.exe --hardware-discover
```

Discovery checks at most 256 HID interfaces until the first exact G13 match. It reports that candidate plus earlier rejection reasons, omits device paths, performs no writes, and does not inspect competing processes. Direct HID

checks both exact `LCore.exe` and `logi_lamparray_service.AMD64.exe` before open, after open, and before each write. It additionally checks `LCore.exe` while idle; LampArray is not polled while idle. Both processes must be absent, and direct HID fails closed if process enumeration fails.

Prefer Logitech Gaming Software (LGS). Start it normally, leave `logitech_backend auto`, and test its SDK connection with:

```
.\LCDSirPlus.exe --hardware-test --backend sdk --duration-secs 60
```

Use direct access only if LGS is not suitable and `--hardware-discover` reports a compatible interface. Exit LCDSirPlus and LGS normally. A direct test is refused while `LCore.exe` or `logi_lamparray_service.AMD64.exe` is running, and it is also refused if another LCDSirPlus process is already running.

If exact `logi_lamparray_service.AMD64.exe` is running, press **Win+R**, run `services.msc`, and find only **Logitech LampArray Service**. Record its current status and startup type, stop that exact service for the test, and do not change its startup type. Do not stop the service when that exact process is not running, and do not stop any other Logitech or G HUB service. Then run the timed direct test:

```
.\LCDSirPlus.exe --hardware-test --backend hid --duration-secs 60
```

After the test, restore the recorded prior service state and confirm its startup type is unchanged.

Watch all ten steps and press all four LCD buttons. Command success means writes completed; still inspect the physical image. Normal `logitech_backend auto` selects the SDK when validated exact `LCore.exe` is present and otherwise tries direct access. Explicit `sdk` and `hid` do not switch to one another.

## Fix common problems

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### Settings do not load

Run `.\LCDSirPlus.exe --validate-config`. Fix the named key and line. A final range or setting combination error can report line `0`. An invalid save leaves the last valid settings active.

### Preview does not appear

Left-click the tray icon. Run `.\LCDSirPlus.exe --preview`, or set `preview_mode always` and `start_minimized 0`. In `auto`, a working G13 normally hides the preview.

### PresentMon shows no game data

The planned packages place `PresentMon.exe` beside `LCDSirPlus.exe`. Confirm that layout, start LCDSirPlus before the game, and select an FPS button option. Set `presentmon_deferred 0` temporarily to keep the unavailable FPS panel visible. If capture is denied, add your user to **Performance Log Users**, sign out, and sign in; use elevation only to diagnose the access problem.

### CPU temperature is N/A or STALE

Repeat the HWiNFO or LibreHardwareMonitor setup above. Use your own exact HWiNFO labels or LibreHardwareMonitor sensor path. Remember the HWiNFO64 Free 12-hour limit. Do not trust a stale temperature.

## Discord speakers do not appear

Confirm the Application ID, tester access, active-account authorization, voice channel, and `discord_show_self` choice. Discord Desktop must run under the same Windows account. Never share credential files.

## Update or uninstall

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When an update is published, exit LCDSirPlus, verify the new setup checksum, and run setup in the same current-user or all-users mode. Running the same setup repairs installed files. Remove an opposite-scope installation owned by the same account before changing scope.

Uninstall from Windows **Installed apps**. The uninstaller removes program files, shortcuts, and its owned sign-in task. It preserves `%LOCALAPPDATA%\LCDSirPlus`, including settings, logs, and Discord credentials. Delete that folder only when its data is no longer needed. Revoke Discord access separately.

For portable removal, set `start_at_login 0`, run that exact portable copy once, exit it, then remove its folder.